



NCCER Module 22107



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Vertical-Mast Sit-Down Counterbalance Forklifts





Section Three

Instrumentation and Controls

Objective

3. Identify and describe instruments and controls found on SDCB forklifts.
 - a. Explain the function of the instruments and controls found in the operator's cab.
 - b. Describe the categories of forklift controls.
 - c. Describe how switches are used on SDCB forklifts.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

2. Demonstrate knowledge of forklift controls found in the operator's cab.



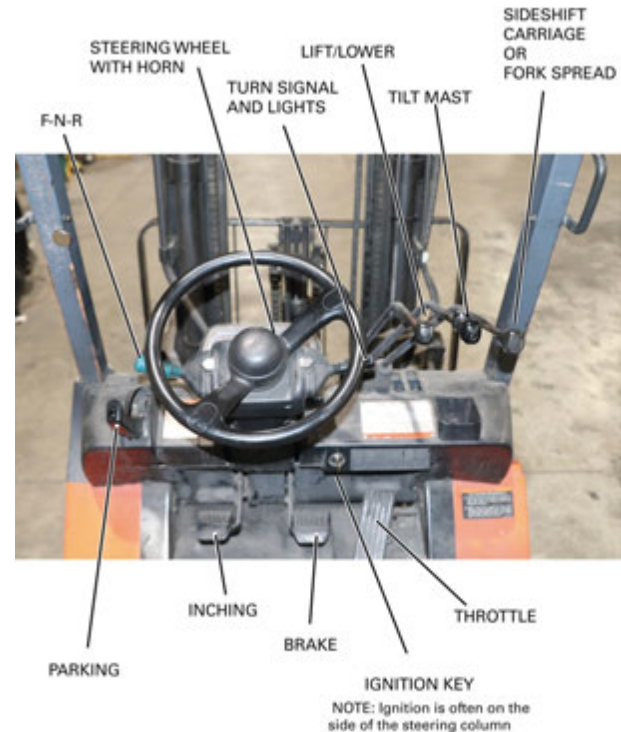
3.0.0 Instrumentation and Controls

- Forklift instrumentation includes gauges, signals, and other indicators used to ensure safe and efficient operation of the machine.
- Forklift controls are used to maneuver the forklift and load and are accessible within the forklift's operator cab.



3.1.0 Operator's Cab

Study the operator's manual to become familiar with functions and features specific to each machine.



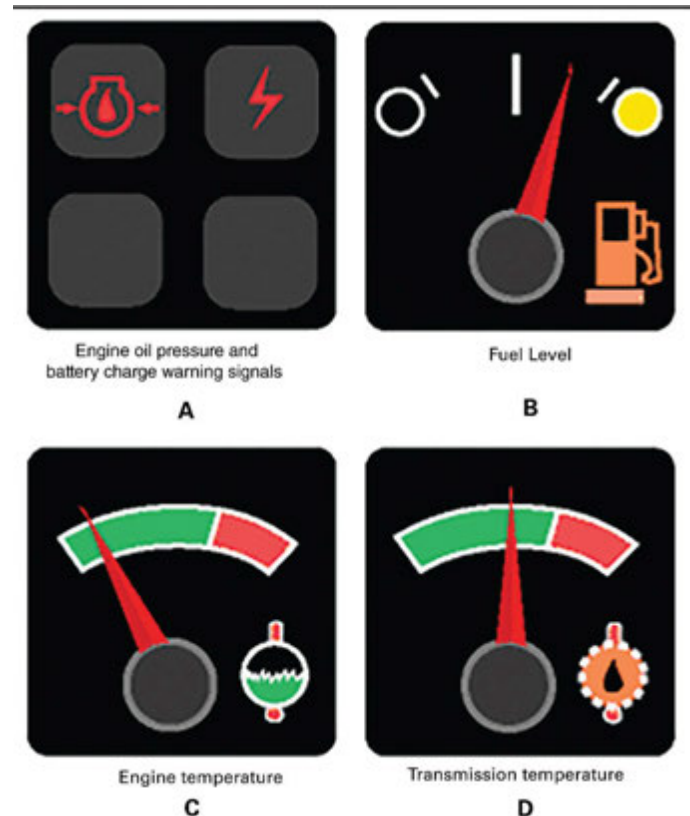
3.1.1 Instrument Panels

- Instrument panels include gauges that monitor engine and transmission temperature, oil pressure, fuel level, and battery charge.
- Panels may be located behind the steering wheel, on the forklift's dash, or as part of the steering column.
- The instrument panel may be a standalone unit.



3.1.2 Indicator Signals and Gauges (1 of 2)

- **Fuel-Level Gauge** – On diesel engine forklifts there may also be a low-fuel warning zone or light.
- **Engine Temperature Gauge** – Indicates the temperature of the coolant. If it gets too high, stop operations immediately.
- **Transmission Oil Temperature Gauge** – The red zone indicates excessively high temperatures.



3.1.2 Indicator Signals and Gauges (2 of 2)



- **Battery Charge Indicator** – Indicates the amount of charge remaining in the forklift's main batteries. Attach the forklift to a charger when the needle enters the red zone.
- **Service Hour Meter** – Indicates the period of time the machine has been running during the life of the forklift. Maintenance should be scheduled and performed based on the time shown on the meter.

3.2.0 Controls

Controls can be grouped into two categories:

- Vehicle Movement
- Mast, Carriage, and Attachment Movement



3.2.1 Vehicle Movement (1 of 3)

- **Forward/Neutral/Reverse (F-N-R) Lever** – This type of lever is used on machines that have a **torque converter drive**, which is a drive commonly used to power forklifts.
- The other form of drive is a **hydrostatic drive**, where the lever controls the flow of hydraulic fluid.



3.2.1 Vehicle Movement (2 of 3)

- A similar type of shifter may be found on electric-powered forklifts.
- It is part of the lift/lower lever and only includes options for forward and reverse.



NOTE: DECAL SHOWS FUNCTION FOR THE LEVER WITH AND WITHOUT THE BUTTON PUSHED

3.2.1 Vehicle Movement (3 of 3)

- **Foot Pedals** – IC-powered forklifts have a throttle, a brake, and an **inching pedal**. Battery-powered forklifts often include only a throttle and a brake pedal.
- **Parking Brake** – Engage the parking brake whenever exiting the forklift or leaving it unattended. A hydrostatic drive forklift typically has a switch that prevents the controls from supplying hydraulic fluid to the drive.



3.2.2 Mast, Carriage, and Attachment Movement (1 of 2)

- The mast and forks are operated with two to four levers that control hydraulic circuits.
- The symbols show the fork's action based on the direction the lever is moved.
- The hydraulic mast controls on a conventional forklift may include a lift cylinder, tilt cylinder, and, depending on the forklift's features, a carriage sideshift and/or forks spread cylinders.



3.2.2 Mast, Carriage, and Attachment Movement (2 of 2)

- **First Lever (lift control)** – This lever controls the lift cylinder to raise and lower the carriage.
- **Second Lever (tilt control)** – This lever controls the tilt of the mast.
- **Third or fourth lever (side shift or spread control)** – These levers control the side shift movement of the carriage or the spreading and narrowing of the forks. They may not be included on some forklift models.



3.2.3 Operator Comfort

- Adjust the seat height, along with the armrests and mirrors, before starting the forklift.
- Adjustments can affect safety as well as operator comfort.
- If you will be working at night, familiarize yourself with the light switches.
- Make sure that all lights are functioning properly and adjust the lights so that the work area is properly illuminated.



3.3.0 Switches

- A switch either makes or breaks an electrical circuit.
- Forklift switches are mechanical or electrical and typically connect or disconnect critical functions such as the starter and lights.



3.3.1 Fuel and Battery Cutoff

- Some machines have fuel shutoff switches that must be turned on before the forklift can be operated. They physically prevent fuel from flowing from the fuel tank into the supply lines.
- Some machines have a battery disconnect switch that disables the entire electrical system. This switch should be turned off when the machine is idle overnight (or longer).



3.3.2 Engine Start Switch

Ignition switch functions can vary between makes and models of forklifts. Most IC-powered forklifts have a multipurpose ignition switch.



The IC-engine start switch has three positions:

- **Off (1)** – Stops the engine and disconnects power to electrical switches and circuits in the cab.
- **On (2)** – Activates all electrical circuits except the starter motor circuit.
- **Start (3)** – Activates the starter, which starts the engine.

3.3.3 Light Switches

- Depending on the forklift's make and model, the light switches may remain illuminated until they are turned off.
- As a result, it is possible to leave the lights on and run down the battery.
- Some machines do not allow the lights to be turned on unless the ignition switch is turned on.



Wrap Up – Trade Terms (1 of 2)

Fork Spread: A forklift feature that hydraulically or manually spreads or narrows the distance between the forks.

Hydrostatic Drive: Hydraulic drive system sometimes used on IC-powered forklifts because it responds quickly and offers seamless, smooth acceleration, regardless of the load. This type of drive does not use a transmission.

Inching Pedal: Braking pedal that initially disengages the transmission before the brake is applied; it works a lot like a clutch with a manual transmission.



Wrap Up – Trade Terms (2 of 2)

Service Hour Meter: Gauge that indicates the period of time the machine has been running during the life of the forklift.

Torque Converter Drive: The type of drive that uses an automatic transmission to drive the wheels of the forklift. A torque converter drive usually has a single-speed, Forward-Neutral-Reverse transmission. It does not have a Park position but can roll freely if it is in neutral.



3.0.0 Section Review Question 1

1. What is the purpose of the two brake pedals found on conventional IC-powered forklifts?
 - a. One brake pedal stops the right front wheel; the other stops the left front wheel.
 - b. One brake pedal is used for regular braking; the other is used to lock the brakes when parked or unattended.
 - c. One brake pedal is used for regular braking; the other is used to disengage the transmission, then brake.
 - d. One brake pedal stops the front wheels; the other stops the back wheels.



3.0.0 Section Review Question 1 Answer

1. What is the purpose of the two brake pedals found on conventional IC-powered forklifts?
 - a. One brake pedal stops the right front wheel; the other stops the left front wheel.
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 - c. One brake pedal is used for regular braking; the other is used to disengage the transmission, then brake.**
 - d. One brake pedal stops the front wheels; the other stops the back wheels.



3.0.0 Section Review Question 2

2. What are the categories of movement controls found in the operator's cab?
- a. Vehicle and Mast
 - b. Hydraulic and Motor
 - c. Direction and Lift
 - d. Levers and Pedals



3.0.0 Section Review Question 2 Answer

2. What are the categories of movement controls found in the operator's cab?

- a. Vehicle and Mast**
- b. Hydraulic and Motor
- c. Direction and Lift
- d. Levers and Pedals



3.0.0 Section Review Question 3

3. Forklift switches can be mechanical or electrical and typically _____ critical functions.
- a. monitor and maintain
 - b. initiate and sustain
 - c. connect or disconnect
 - d. assist and support



3.0.0 Section Review Question 3 Answer

3. Forklift switches can be mechanical or electrical and typically _____ critical functions.
- a. monitor and maintain
 - b. initiate and sustain
 - c. connect or disconnect**
 - d. assist and support



3.0.0 Section Review Question 4

4. True or False: Leaving the ignition key in the On position without the engine running can drain the machine's battery.
- a. True
 - b. False



3.0.0 Section Review Question 4 Answer

4. True or False: Leaving the ignition key in the On position without the engine running can drain the machine's battery.
- a. True**
 - b. False



Next...

Section 4.0.0
Forklift Attachments



*Read Sections 4.0.0 through 4.2.6 and
complete the Section Review.*